CSC 4500 Database Systems

HW 2 – Due Oct 12, 2026 (Monday), 11:59 PM; 85 points

Submit the following as .TXT and .PDF files in Brightspace:

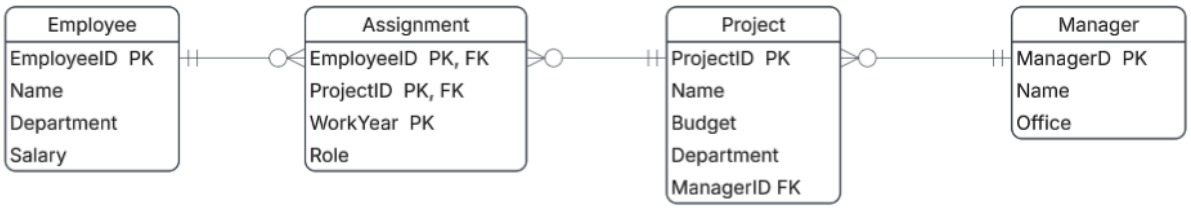
1. **SQL (.txt)**
2. **Screenshots of the results (.pdf)**

This is an individual assignment.

HW 2 p2 will be in class on 10/13 (Tuesday) to write one query in BS Quiz; 15 points

SQL – DDL, DML, and DQL Implementations

1. Using this ERD, create the appropriate SQL DDL implementation, including all tables, relationships, cardinalities, and ordinalities.



2. Insert the values (SQL DML) into the tables as shown below. Remember that some columns may need to be modified so they can be used as the appropriate foreign keys.

Employee

ID	Name	Department	Salary
1001	Daniel Smith	Accounting	62,000.00
1002	Maria Johnson	Information Technology	78,000.00
1003	Kevin Taylor	Human Resources	58,000.00
1004	Sophia Lee	Information Technology	85,000.00
1005	Emma Garcia	Marketing	67,000.00
1006	Michael Wilson	Accounting	72,000.00
1007	Henry Brown	Human Resources	61,000.00
1008	James Anderson	Information Technology	74,000.00
1009	Grace Martinez	Marketing	69,000.00
1010	John Thomas	Information Technology	92,000.00
1011	Anna Davis	Accounting	55,000.00
1012	Leo Hernandez		60,000.00

Assignment

Employee	Project	WorkYear	Role
Daniel Smith	Financial Audit	2024	Analyst
Maria Johnson	Payroll System Upgrade	2026	Developer
Kevin Taylor	Employee Wellness Program	2026	Coordinator
Sophia Lee	Network Security Upgrade	2025	Developer
Emma Garcia	Financial Audit	2026	Designer
Michael Wilson	Financial Audit	2024	Analyst
Henry Brown	Employee Wellness Program	2026	Coordinator
James Anderson	Payroll System Upgrade	2026	Developer
Grace Martinez	Financial Audit	2026	Designer
John Thomas	Network Security Upgrade	2025	Developer
Anna Davis	Financial Audit	2024	Analyst
Leo Hernandez	Payroll System Upgrade	2026	Support

Project

ID	Name	Budget	Department	Manager
PRJ1001	Payroll System Upgrade	85,000.00	Information Technology	Robert Miller
PRJ1002	Network Security Upgrade	120,000.00	Information Technology	Robert Miller
PRJ2001	Employee Wellness Program	45,000.00	Human Resources	Susan Clark
PRJ3001	Marketing Campaign	70,000.00	Marketing	Michael Chen
PRJ4001	Financial Audit	50,000.00	Accounting	Michael Chen

Manager

ID	Name	Office
201	Robert Miller	ROOM 201
202	Susan Clark	ROOM 305
203	Michael Chen	ROOM 410

3. Show all projects that have no employees assigned to them. Use 'NoEmployeeProject' as the column alias for the project name.
4. Using JOIN, show the employee ID, name, and department for all employees assigned to a project in the Information Technology department.
5. Show the project ID and project name for the project(s) with the most employees assigned — using a subquery.
6. Show the project ID and project name for the project(s) with the most employees assigned — using a CTE.
7. Show the manager's ID, Manager's Name, total project budget managed by that manager, and the average total project budget managed by all managers.